

DETROIT LAKES, MN, USA

WMO: 727457

Lat: 46.833N

Lon: 95.883W

Elev: 1397

StdP: 13.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-14.9	-26.5	1.3	-19.6	-20.3	1.9	-13.2	28.0	8.9	25.8	9.8	5.5	330	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	87.6	71.7	82.4	68.3	81.1	67.3	74.2	82.8	72.0	80.0	70.0	77.5	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	124.5	79.6	69.7	115.2	77.9	67.6	106.7	75.7	38.7	83.4	36.7	79.8	34.8	77.6	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.6	19.2	-25.7	92.5	7.3	3.4	-31.0	94.9	-35.2	96.8	-39.3	98.7	-44.6	101.1
			-26.1	77.9	7.1	3.2	-31.2	80.2	-35.4	82.0	-39.4	83.8	-44.6	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	9.8	13.7	27.2	42.1	55.0	65.2	70.0	67.9	59.5	45.5	30.4	16.5
	DBStd	23.44	14.69	13.48	13.01	10.16	8.68	6.68	5.59	5.68	8.15	8.93	11.08	12.75
	HDD50	5137	1247	1016	711	273	49	1	0	0	14	198	590	1038
	HDD65	8818	1712	1436	1173	688	325	77	20	35	203	608	1038	1503
	CDD50	2231	0	0	3	35	204	457	619	555	298	57	3	0
	CDD65	435	0	0	0	0	16	83	173	124	37	2	0	0
	CDH74	3271	0	0	0	14	172	631	1280	877	276	21	0	0
	CDH80	749	0	0	0	2	34	139	317	198	57	2	0	0
Wind	WSAvg	8.7	8.8	8.7	9.2	9.5	9.5	8.3	7.3	7.4	8.6	9.7	9.4	8.7
Precipitation	PrecAvg	26.1	0.7	0.6	1.1	1.9	3.1	4.4	3.8	3.4	3.0	2.3	0.9	0.7
	PrecMax	36.8	2.6	2.0	2.7	5.2	8.8	9.3	11.1	7.9	9.9	6.3	3.9	1.9
	PrecMin	12.6	0.1	0.0	0.1	0.1	0.4	1.2	0.4	0.3	0.4	0.1	0.0	0.1
	PrecStd	5.4	0.5	0.4	0.6	1.3	1.9	2.0	2.0	1.8	1.9	1.8	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	45.4	63.3	75.0	84.1	89.8	90.5	90.4	86.6	77.0	62.5	44.1
		MCWB	36.5	41.1	53.2	56.0	63.0	72.3	73.8	72.7	69.4	60.9	52.4	38.8
	2%	DB	36.0	39.4	54.8	69.5	79.0	83.5	87.7	84.4	81.2	70.0	54.7	38.6
		MCWB	33.5	36.2	47.0	52.7	61.5	69.1	73.1	69.7	66.9	57.7	46.5	35.1
	5%	DB	32.3	36.3	49.5	63.2	73.4	80.8	82.4	81.5	77.1	63.6	50.0	35.9
		MCWB	31.4	33.9	43.6	48.8	57.9	65.9	69.5	67.9	64.6	54.2	44.0	33.5
	10%	DB	30.1	32.4	43.2	59.0	70.1	77.4	81.0	79.2	72.8	59.1	45.1	32.3
		MCWB	29.4	31.2	38.3	46.4	56.9	63.8	68.4	66.6	62.8	50.7	40.2	31.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	41.7	55.0	58.1	68.2	75.3	77.4	76.2	72.8	63.3	53.8	39.6
		MCDB	40.9	46.2	59.0	72.4	78.2	84.9	85.5	84.7	81.2	70.9	59.1	43.5
	2%	WB	33.6	37.0	47.9	53.4	64.3	71.8	75.0	73.0	69.5	59.2	48.0	35.8
		MCDB	35.3	39.9	54.1	63.8	73.8	81.0	83.6	80.9	76.7	67.5	53.9	38.7
	5%	WB	31.2	34.0	43.2	50.5	61.3	68.9	72.7	70.8	67.2	55.3	43.9	33.2
		MCDB	31.9	36.0	48.5	61.4	70.1	77.0	80.6	78.0	74.6	62.8	49.3	34.8
	10%	WB	29.0	31.2	38.9	47.7	58.6	66.6	70.5	68.8	64.5	52.1	40.0	31.3
		MCDB	29.8	32.4	43.2	57.3	67.5	74.3	78.1	76.1	70.9	57.5	44.7	32.3
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.1	16.1	18.1	18.9	17.3	17.2	17.6	17.5	14.7	12.9	14.0
		MCDBR	15.8	17.2	22.2	26.0	25.1	20.9	19.4	20.3	22.0	22.5	19.6	15.3
		MCWBR	13.9	14.2	14.5	13.8	12.5	10.2	9.7	9.6	11.0	12.7	12.7	12.3
	5% WB	MCDBR	14.8	16.0	21.4	24.1	21.6	17.9	18.2	18.5	18.8	20.2	18.6	14.2
		MCWBR	13.3	13.4	14.6	14.0	12.5	10.7	10.9	10.6	11.3	12.7	12.8	12.1
Clear-Sky Solar Irradiance	taub	0.276	0.303	0.297	0.350	0.366	0.376	0.378	0.381	0.348	0.306	0.279	0.260	
	taud	2.303	2.229	2.421	2.353	2.370	2.372	2.376	2.381	2.455	2.554	2.521	2.406	
	Ebn at Noon	258	272	296	287	285	282	279	274	274	273	261	255	
	Edh at Noon	24	32	31	36	37	37	37	35	30	23	19	19	
All-Sky Solar Radiation	RadAvg	514	875	1220	1543	1718	1890	2000	1682	1253	775	476	360	
	RadStd	56	90	123	130	117	127	86	140	89	116	51	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page